



RESUME SEARCH AND CANDIDATE SCREENING SYSTEM. TALENTFLOW

**An Advanced Algorithm-Based Resume Search and
Candidate Screening Engine**

Engineering the Future of Intelligent Recruitment

YASHAS KASULA - 2520030401

GNANESHWAR REDDY - 2520030552

CHARAN SAI - 2520030321

SECTION - 8 TEAM - 6

Abstract



Precision search

Combine exact skill matching with advanced pattern and fuzzy algorithms to surface the right candidates within seconds.

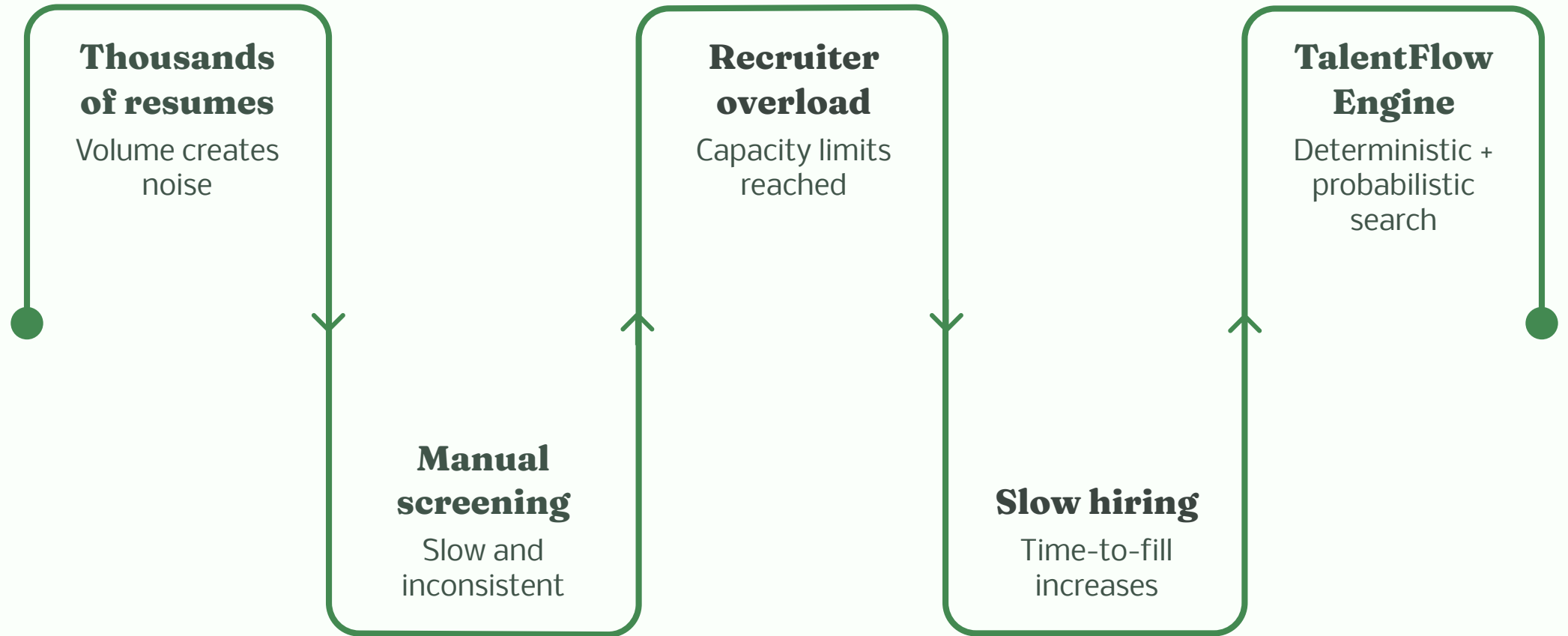
Scale & speed

Process thousands of resumes with near-real-time indexing and typo-tolerant retrieval – designed for enterprise throughput.

Actionable outcomes

Integrated ranking, interview assignment, and analytics to shorten time-to-hire and improve quality of hire.

Problem Statement



Current pipelines are overwhelmed by volume and noise. TalentFlow replaces manual bottlenecks with deterministic and probabilistic search that elevates relevant candidates fast.

Proposed Functionalities



Resume Parsing

Robust extraction of skills, roles, dates and entities from diverse formats (PDF, DOCX).



Exact Skill Search

Deterministic matching for role-specific skillsets and certifications.



Multi Pattern Search

Complex multi-field queries: experience patterns, project types, technology stacks.



Fuzzy Skill Matching

Typo-tolerant and synonym-aware retrieval preserves recall without sacrificing precision.



Resume Similarity Analysis

Cluster and dedupe candidates; find near-matches and internal mobility opportunities.



Candidate Ranking

Customisable scoring model with weighting for skills, seniority and culture fit signals.



Interview Assignment

Automated interviewer pairing and calendar invitation workflows to reduce coordination time.



Analytics Dashboard

Real-time KPIs: search latency, conversion funnels, top skill demand and diversity metrics.

Technology Stack

Infrastructure & Integrations

Java
Proven backend runtime for high-throughput services.

Java Swing / JavaFX
Premium internal tools and rapid desktop QA utilities.

Apache PDFBox
Reliable parsing and text extraction from PDFs.

MySQL & JDBC
Durable relational storage for metadata and audit logs.

Git / GitHub
Enterprise CI/CD, code review and secure deployment pipelines.

Core Algorithms & Theory

Knuth–Morris–Pratt
Deterministic substring search for exact phrase matching at scale.

Rabin–Karp
Rolling-hash accelerated pattern detection across large collections.

Edit Distance
Levenshtein-based matching for typo-tolerance and fuzzy searches.

Aho–Corasick
Simultaneous multi-pattern matching for skill dictionaries and keywords.

Maximum Bipartite Matching
Optimal assignment for interviewer - candidate pairing and batch scheduling.

Hashing & Complexity Analysis
Scalable indexing, bloom filters and theoretical performance guarantees.

Expected Outcome

90%

Faster Resume Search

Median query times reduced by an order of magnitude versus manual search.

High

Search Accuracy

Precision-focused retrieval with configurable recall/precision trade-offs.

Typo-Tol...

Robust Matching

Handles misspellings, abbreviations and role variations reliably.

Automated Interview Scheduling

Seamless calendar integration and interviewer allocation to cut coordination overhead.

Scalable Candidate Search

Designed for millions of indexed profiles with horizontal scaling patterns.

Algorithm Performance Analysis

Built-in benchmarking and profiling to continuously validate speed and accuracy.

Applying Advanced Algorithms to Transform Modern Recruitment.

THANK YOU